

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

At birth , the fetal brain undergoes a “pruning” and redistribution of neurons and synapses in a largely reversible manner . The primary goal is to streamline neural connections to meet the demands of daily life and rapidly acquired perceptual/motor skills , while conserving resources . This process includes :

- Synaptic pruning : excess excitatory and inhibitory connections are removed .
- Detruncation of axons : fibers are shortened so the brain does not expand excessively upon physical maturation .
- Functional refinement : development of higher-order networks (such as auditory and visual pathways) occurs later than basic functions like somatosensory integration and reflexes , and is better preserved .

These changes are most readily evident in the visual system . In humans , most early visual processing develops at birth ; although retinal ganglion cell axons and optic nerve connections are still incomplete (eg , “white matter” delay) , functional perception “such as basic vision of moving objects and edges” manifests early . In contrast , more delayed connections include complex visual pathways (e.g . , V1 thalamocortical projections) , auditory pathways , and advanced motor planning .

Thus , at birth the least developed functional system is the visual system , and the limbic system is the least developed anatomically by birth , not at birth .

Correct answer : A . visual system

</think>

<answer>

A . visual system

</answer>